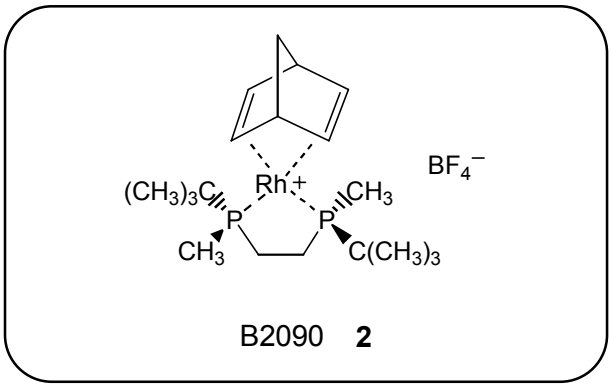
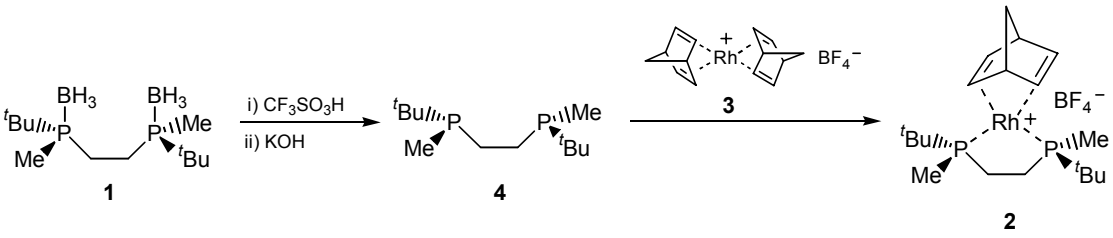
Highly Enantioselective Hydrogenation and Hydrosilylation Reactions

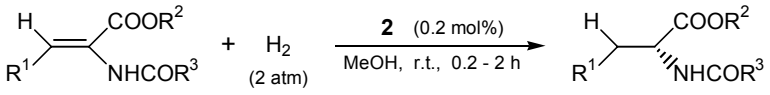


B2090 **2**

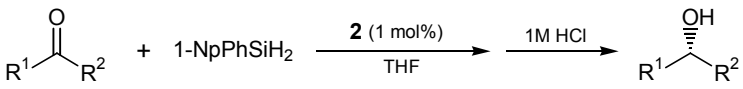
While catalytic asymmetric reactions using optically active phosphine ligands represented by BINAP, have been vigorously investigated, the majority of the phosphine ligands employed in these reactions bear C-chirality or axial asymmetry, and only a limited number of P-chiral phosphines have been reported.[1)] Imamoto and co-workers developed a synthetic method for optically pure phosphine-borane **1**, and **1** was treated with strong-acid to yield P-chiral phosphine ligand **4** without racemization. They reported that the Rh Complex **2** prepared from **4** showed high catalytic activity and enantioselectivities for the hydrogenation of dehydroamino acid derivatives.[2,3)]

Hereafter, further application of **2** and **4** is anticipated to a wide variety of catalytic asymmetric reactions.[4,5)]





R^1=Ph, R^2=Me, R^3=Me 99.9% *e.e.*
R^1=Ph, R^2=H, R^3=Me 98.4% *e.e.*
R^1=3-MeO-4-AcOC_6H_3, R^2=Me, R^3=Me 99.8% *e.e.*



R^1=2-MeOPh, R^2=Me 99% *e.e.*(*R*)
R^1=2,4-Cl_2Ph, R^2=CH_2Cl 99% *e.e.*(*S*)

Keywords *: enantioselective hydrogenation, hydrosilylation*

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Product

B2090	(*S*,*S*)-1,2-Bis[(*tert*-butyl)methylphosphino]ethane[η-(2,5-norbornadiene)]-rhodium(I) Tetrafluoroborate	(**2**)	50mg

Related Products

B2089	(*S*,*S*)-1,2-Bis[(*tert*-butyl)methylphosphino]ethane Bis(borane)	(**1**)	100mg
B2091	Bis[η-(2,5-norbornadiene)]rhodium(I) Tetrafluoroborate	(**3**)	100mg

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